Engineering - Undergraduate Level Assessment 4 Total Marks: 40 Answer all questions.

Multiple Choice (10 marks)

1. The rating of a certain machine from the name-plate is 110 volts, 38.5 amperes, 5 h.p. Find the input, output and efficiency at full load.

   (a) Input: 4235 watts, Output: 3600 watts, Efficiency: 85 percent
   (b) Input: 4500 watts, Output: 4000 watts, Efficiency: 88.9 percent
   (c) Input: 4350 watts, Output: 3730 watts, Efficiency: 85.7 percent
   (d) Input: 4100 watts, Output: 3450 watts, Efficiency: 84.1 percent
   (e) Input: 4200 watts, Output: 3900 watts, Efficiency: 86 percent

2. Evaluate the determinant | 1- 4- 5| Δ=| 123| |- 31- 2|

   (a) -28
   (b) 28
   (c) -7
   (d) 0
   (e) 7

3. Electrons in p-type material of a semi-conductor are called as

   (a) majority carriers
   (b) dominant carriers
   (c) neutral carriers
   (d) non-conductive carriers
   (e) charge carriers

4. The feedback factor of a Wien bridge oscillator using Op-Amp is

   (a) 3/2
   (b) 1/4
   (c) 0
   (d) 1/5
   (e) 1/3

5. The field winding of a shunt motor has a resistance of 110 ohms, and the voltage applied to it is 220 volts. What is the amount of power expended in the field excitation ?

   (a) 660 watts
   (b) 880 watts

(c) 200 watts

(d) 120 watts

(e) 110 watts

True / False (10 marks)

Answer True or False

6. True or False: The discharge from a rectangular broad crested weir with a 2 ft head and a discharge coefficient of 0.7 is 476 cfs.

7. True or False: The time function corresponding to F(s) = [(s^2 + 3 s + 1) / {(s + 1)^3 (s + 2)^2}] is F(t) = - t^2 e$^-t + 4te^-t - 5e^-t + 2te^-2t + 5e^-2t$.

7. True or False: Power dissipation in an ideal inductor is maximum, indicating significant energy loss during operation.

8. True or False: The potential energy of an isolated spherical conductor of radius R with surface charge density is Q^2 / (8R).

9. True or False: The theoretical variance of the given probability distribution is 1.125.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the factors influencing the internal pressure of a water droplet at $20^a nddeterminethenecessary$

11. Explain how to derive the group velocity of a plane wave in a normally dispersive, lossless medium, given that its phase velocity is expressed as v = k$\sqrt{}$, where k is a constant.

End of Paper